step $x > 0 \rightarrow 1$.
$x < 0 \rightarrow 0$

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Perception → only mis classified point
line-pull

new approach ⇒ correctly classified point.
↳ line-push.

both. ⇒ mis classified point
line-pull

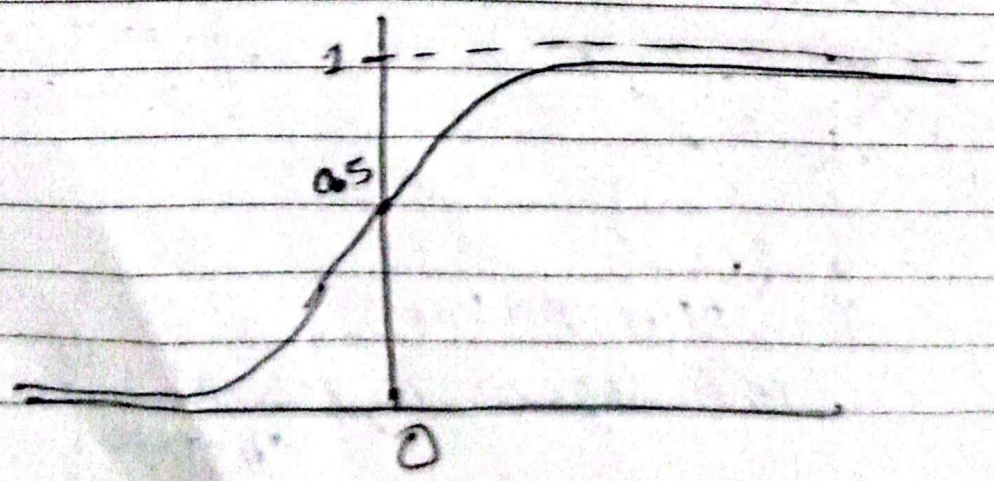
- How much force does point exert in the line also depend in distance betn line & point.

nearer → ~~more~~ less force
farther → ~~less~~ more force.
} if misclassified

nearer → ~~less~~ more force
farther → ~~more~~ less force
} if correct classified.

Sigmoid Function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$



$\frac{1}{1+e^{-z}} \rightarrow 1/(1+np.exp(-z))$

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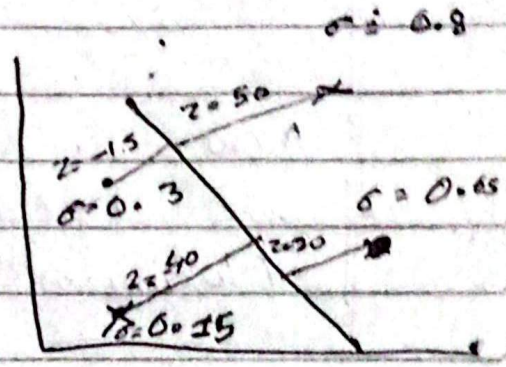
$\sum w_i x_i = z \rightarrow \sigma(z) = \frac{1}{1+e^{-z}}$

If $z \geq 0 \rightarrow \sigma \geq 0.5 \rightarrow 1$
If $z < 0 \rightarrow \sigma < 0.5 \rightarrow 0$

If point lies on point $\rightarrow$
$\sum wx = 0 \rightarrow \sigma(z) \rightarrow 0.5$.

- It also represent Probability

Impact of Sigmoid f^n



$w_i = w_0 + \eta (y_i - \hat{y}_i) x_i$

$\hat{y}_i = \sigma(z) = \frac{1}{1+e^{-z}}$

where $z = \sum w_i x_i$

y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	0.8	0.2
0	0.65	-0.65
1	0.3	0.7
0	0.15	-0.15

Here, unlike step [always 1 or 0] diff values of $y_i - \hat{y}_i$

So; $w_i = w_0 + \eta (y_i - \hat{y}_i) x_i$ will be always corrected either positive or negative.

eg: $w_1 = w_0 + \eta \times 0.2 \times x_1$

party.

x → yes
• → no

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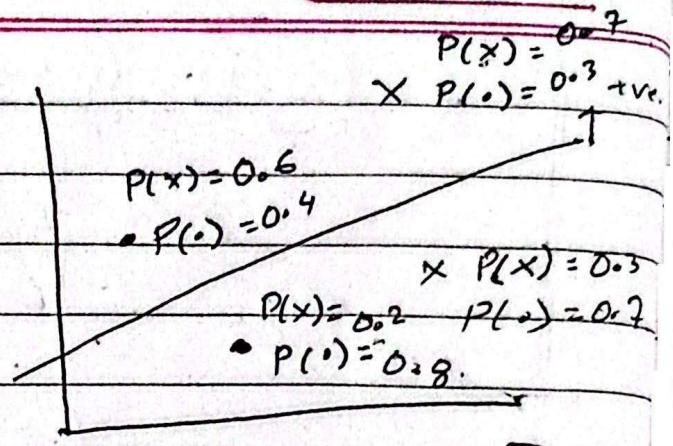
Maximum Likelyihood.



~~What~~
watch • or x → use Probability according to its shape.

model • → $P(x) \times P(x) \times P(\cdot) \times P(\cdot)$

Max^m Likelihood. = $0.7 \times 0.3 \times 0.8 \times 0.4$

Higher the above multiplication, better will be the model.

Now;

- we need a line in which Maximum Likelihood value will be higher.
- It is better to calculate the better model.
  But in real world with (100000 rows) higher rows → the value of Maximum Likelihood will be ~~to~~ too much lower. so, diff to compare.

So; we take the log of above Maximum Likelihood value.

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$= \log(0.7 \times 0.3 \times 0.8 \times 0.4)$

$= \log(0.7) + \log(0.3) + \log(0.8) + \log(0.4)$

But we know, value : 0-1.
then log value : negative
like $\log(0.2) = -0.69$.

So; we do negative

$= -\log[0.7 \times 0.3 \times 0.8 \times 0.4]$

$= -\log(0.7) - \log(0.3) - \log(0.8) - \log(0.4)$

This is Cross Entropy:
↓
~~Submission~~ Σ of negative log of maximum likelihood.

- Here, we have to find the minimum value of Cross Entropy:
  coz: $-\log(0.1) > -\log(0.9)$
  $-(-1) > -(-0.045)$

Here, we can't generalize formula lik-
$Loss = -\log(\hat{y}_1) - \log(\hat{y}_2) - \log(\hat{y}_3) - \ldots$

coz $\{\hat{y}_1, \hat{y}_2, \hat{y}_3\} \rightarrow$ कुन $P(x)$ or use गर्ने $P(\cdot)$
won't be mentioned.
either $(\hat{y}_1)$ or $(1 - \hat{y}_1)$

Then $f^n = -y_i \log(\hat{y}_i) - (1-y_i)\log(1-\hat{y}_i)$

If $\rightarrow y_i = 1$, then

$$-y_i \log(\hat{y}_i) - (1-1)\log(1-\hat{y}_i)$$
$$= -y_i \log(\hat{y}_i)$$

If $\rightarrow y_i = 0$, then

$$= 0 \cdot \log(\hat{y}_i) - (1-0)\cdot\log(1-\hat{y}_i)$$
$$= -\log(1-\hat{y}_i)$$

Then.

Total Loss $f^n = \sum_{i=1}^{n} -y_i \log(\hat{y}_i) - (1-y_i)\log(1-\hat{y}_i)$

As;

Avg. Loss $f^n = -\frac{1}{n} \sum_{i=1}^{n} y_i \log(\hat{y}_1) + (1-y_i)\log(1-\hat{y}_i)$

→ Log-Loss Error

→ Binary Cross Entropy.

Now, we have to find minimum value by gradient descent.

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Derivative of Sigmoid Function

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\Rightarrow \sigma(x) = (1+e^{-x})^{-1}$$

$$\Rightarrow \sigma'(x) = \frac{-1}{(1+e^{-x})^2} \times (-1) \cdot e^{-x}$$

$$\therefore \sigma'(x) = \frac{e^{-x}}{(1+e^{-x})^2}$$

or, $$\sigma'(x) = \frac{1}{1+e^{-x}} \cdot \frac{e^{-x}}{1+e^{-x}}$$

$$\therefore \sigma'(x) = \frac{\sigma\{e^{-x}+1-1\}}{1+e^{-x}}$$

or, $$\sigma'(x) = \sigma \cdot \left[\frac{e^{-x}+1}{e^{-x}+1} - \frac{1}{1+e^{-x}}\right]$$

or, $$\sigma'(x) = \sigma \cdot [1-\sigma)$$

$$\therefore \sigma'(x) = \sigma(x)[1-\sigma(x)]$$

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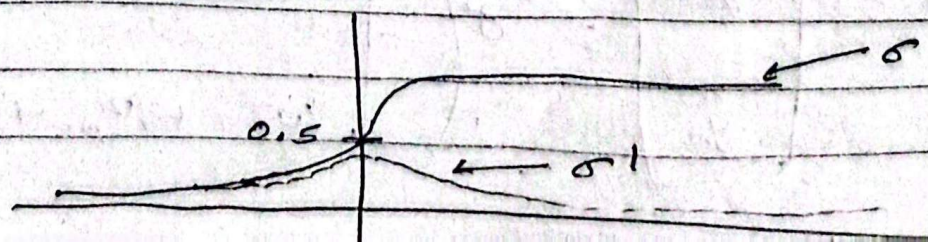
$$\boxed{\therefore \sigma'(x) = \sigma(1-\sigma)}$$